Candy Sales Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 10,194 purchases across various candy categories. The goal is to uncover insights into spending patterns, customer segments, candy preferences and behavior to guide strategic business decisions.

2. Dataset Summary -

-Rows: 10,194
- Columns: 28
 - Key Features: - Customer demographics (Order id, Location (Latitude,Longitude), State/Province, City,Country,Region)
- Purchase details (Product id,Product name,sales,units,)
- Factory details (Factory,Unit cost ,Unit Price,Tagets,Profit Margin,Shipment mode)
 - Missing Data: 6 values in Factory,Unit Price ,Unit Cost columns.

3. Exploratory Data Analysis using Python

Data cleaning and preparation through python.
Data Loading : Imported using the *pandas.*

Initial Exploration: used *df.info()* to check structure and *.describe()* for summary statistics.

Data Merging : Merging the datasets of Sales ,Products,Factories,to make master data set Candy Master file for easier data analysis.

Handling Missing Values: There is no nulls in individual Data sets ,6 Null are in Candy Master file . Deleted the Null Values.

Attribute Normalization: Relabeled attributes to **snake case** for improved legibility and reference.

Attribute Creation:
Formed **year** attribute by categorizing order date.
Formed **Profit Margin** attribute from Pecntage of Gross profit ,sales .

- **Data Storage Connection:** Linked Python code to MySQL and imported the purified DataFrame into the storage for SQL examination.

4. Information Examination with SQL (Corporate Operations)

We performed structured analysis in MySQL to answer key business questions:

1. Helps identify **top-selling products** and trends in demand.

product_name	total_sales	total_units
Wonka Bar - Triple Dazzle Caramel	28485	7596
Wonka Bar -Scrumdiddlyumptious	27874.800000000043	7743
Wonka Bar - Milk Chocolate	26867.75	8267
Wonka Bar - Fudge Mallows	24890.400000000012	6914
Wonka Bar - Nutty Crunch Surprise	23574.949999999888	6755
Lickable Wallpaper	7860	393
Kazookles	1205.75	371
Wonka Gum	597.5	478
Everlasting Gobstopper	130	13

2. **Highlights profitable regions** for targeted expansion or resource allocation.

region	year	avg_profit_margin
Interior	2023	67.24891937390714
Interior	2024	67.00276705675833
Pacific	2022	66.83464151042241
Interior	2022	66.7162764442665
Pacific	2021	66.63864409761409
Gulf	2024	66.59182065342532
Pacific	2024	66.57638312623426
Gulf	2022	66.5102847178105
Gulf	2023	66.45376178131562
Atlantic	2024	66.37368917596025
Pacific	2023	66.36733371105038
Atlantic	2023	66.3656413861978

3. Evaluates **supplier performance** to optimize costs and partnerships.

factory	total_gross_profit
Lot's O' Nuts	52771.05000000059

4. Assesses **shipping efficiency** and its impact on sales volume.

ship_mode	num_orders	avg_sales
Standard Class	5138	13.968950294310039
Second Class	1653	14.077928246589186
First Class	1310	13.775478654592478
Same Day	442	13.00488117001827

5. Pinpoints **high-demand urban areas** for localized promotions.

	city	total_sales
▶	Los Angeles	10408.749999999989
	San Francisco	7018.61
	Seattle	5893.389999999997
	San Diego	2348.879999999997
	Phoenix	785.3100000000003

6. Tracks **product demand trends** to forecast future inventory needs.

	product_id	year	total_units	prev_units	growth_pct
▶	CHO-FUD-51000	2021	1449	NULL	NULL
	CHO-FUD-51000	2022	1410	1449	-2.6915
	CHO-FUD-51000	2023	1830	1410	29.7872
	CHO-FUD-51000	2024	2225	1830	21.5847
	CHO-MIL-31000	2021	1757	NULL	NULL
	CHO-MIL-31000	2022	1741	1757	-0.9106
	CHO-MIL-31000	2023	2076	1741	19.2418
	CHO-MIL-31000	2024	2693	2076	29.7206
	CHO-NUT-13000	2021	1259	NULL	NULL
	CHO-NUT-13000	2022	1355	1259	7.6251
	CHO-NUT-13000	2023	1912	1355	41.1070
	CHO-NUT-13000	2024	2229	1912	16.5795

7. Identifies **key customers** for loyalty programs or personalized offers.

customer_id	total_spend	num_order
131807	454.40000000000003	15
115238	380.5	15
122336	352.2	3
164756	304.14	5
124163	247.35	4
164770	232.4	2
125388	216.00000000000003	6
163790	208.95	6
164147	204.64999999999998	3
140326	202.05	3

8. **Detects pricing inconsistencies** for data quality assurance and profit accuracy.

product_name	price_discrepancy
Wonka Bar - Triple Dazzle Caramel	0
Wonka Bar -Scrumdiddlyumptious	0
Wonka Bar - Fudge Mallows	0
Wonka Bar - Milk Chocolate	0
Wonka Bar - Nutty Crunch Surprise	0
Wonka Gum	0
SweeTARTS	0
Lickable Wallpaper	0
Kazookles	0
Everlasting Gobstopper	0
Nerds	0
Fun Dip	0

9. Total sales, total units sold, and average profit margin for each product_name in the 'Pacific' region during the year 2023?

product_name	total_sales	total_units	average_profit_margin
Wonka Bar - Triple Dazzle Caramel	2396.25	639	65.3333333333334
Wonka Bar - Milk Chocolate	2154.75	663	64.92307692307689
Wonka Bar - Fudge Mallows	2044.7999999999993	568	66.66666666666663
Wonka Bar -Scrumdiddlyumptious	1980.000000000001	550	69.44444444444457
Wonka Bar - Nutty Crunch Surprise	1877.6200000000033	538	71.34670487106021
Lickable Wallpaper	860	43	50
Kazookles	113.75	35	7.6923076923076925

5. Exploratory Data Analysis (EDA)

Summary Statistics

Numerical fields in Sales:

Statistic	Sales	Units	Gross Profit	Cost	Profit Margin
count	10194.0	10194.0	10194.0	10194.0	10194.0
mean	13.91	3.79	9.17	4.74	66.51
std	11.34	2.23	6.64	5.06	6.72
min	1.25	1.0	0.25	0.60	7.69
25%	7.20	2.0	4.90	2.40	65.33
50%	10.80	3.0	7.47	3.60	66.67
75%	18.00	5.0	12.25	5.70	69.44
max	260.00	14.0	130.00	130.00	80.00

Average order: $13.91 sales, 3.79 units, $9.17 profit (66.5% margin).

- Variability suggests occasional large orders.

Unique counts:

- Divisions: Chocolate (9236), Other (915), Sugar (43)
- Regions: Pacific (3262), Atlantic (2912), Interior (2261), Gulf (1759)
- Products: 15 unique

Key Aggregations

Total Sales by Division:

Division	Total Sales
Chocolate	131692.90
Other	9663.25
Sugar	427.48

Top 5 Products by Gross Profit:

Product Name	Gross Profit
Wonka Bar -Scrumdiddlyumptious	19357.50
Wonka Bar - Triple Dazzle Caramel	18610.20
Wonka Bar - Milk Chocolate	17443.37
Wonka Bar - Nutty Crunch Surprise	16819.95
Wonka Bar - Fudge Mallows	16593.60

Yearly Sales:

Year	Total Sales
2021	28953.55
2022	29002.65
2023	36859.32
2024	46968.11

- 62% growth from 2021 to 2024.

Sales by Region:

Region	Total Sales
Pacific	46301.53
Atlantic	41197.24
Interior	32037.60
Gulf	22247.26

Average Profit Margin by Division (%):

Division	Margin
Chocolate	67.46
Sugar	57.69
Other	37.67

Sales by Factory:

Factory	Total Sales
Lot's O' Nuts	76340.15
Wicked Choccy's	55352.75
Secret Factory	8587.50
The Other Factory	1282.25
Sugar Shack	142.23

Top 5 Cities by Sales:

City	Total Sales
New York City	12514.90
Los Angeles	10408.75
Philadelphia	7415.52
San Francisco	7018.61
Seattle	5893.39

6. 2024 Performance vs Targets

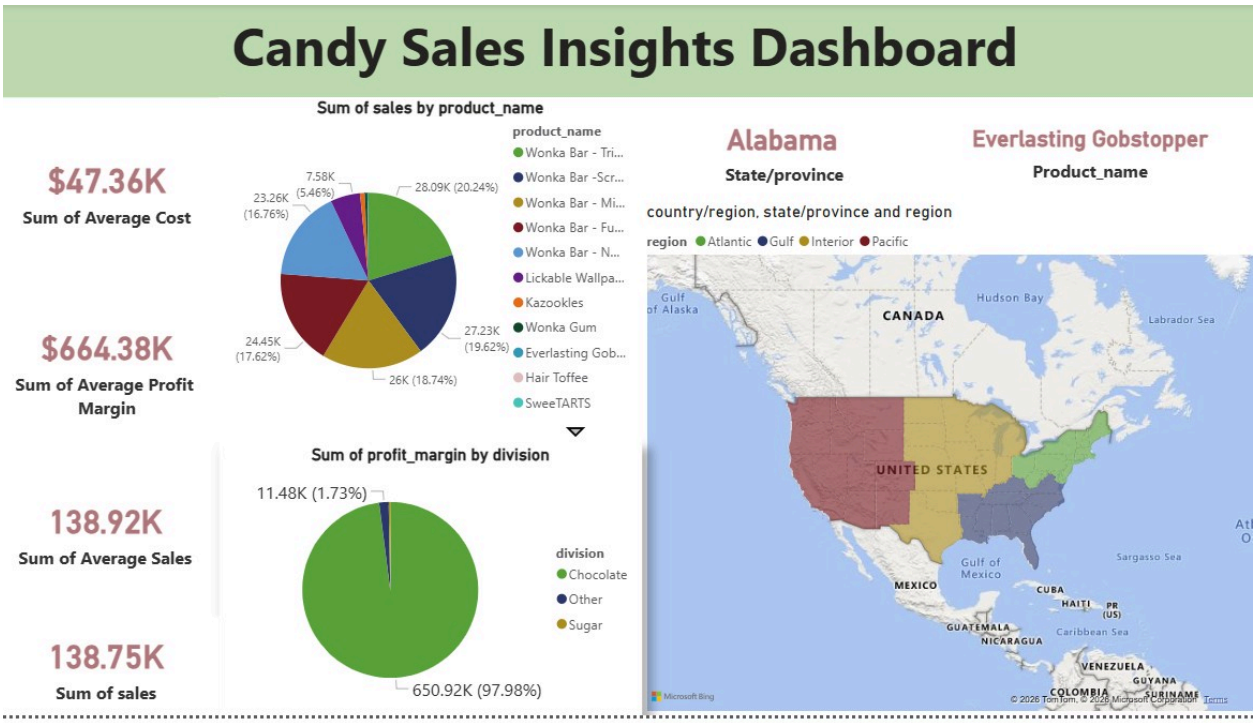
Division	Actual Sales	Target	Variance
Chocolate	43379.96	27000	16379.96
Other	3444.50	3000	444.50
Sugar	143.65	15000	-14856.35

- Chocolate and Other surpassed targets; Sugar fell short dramatically.

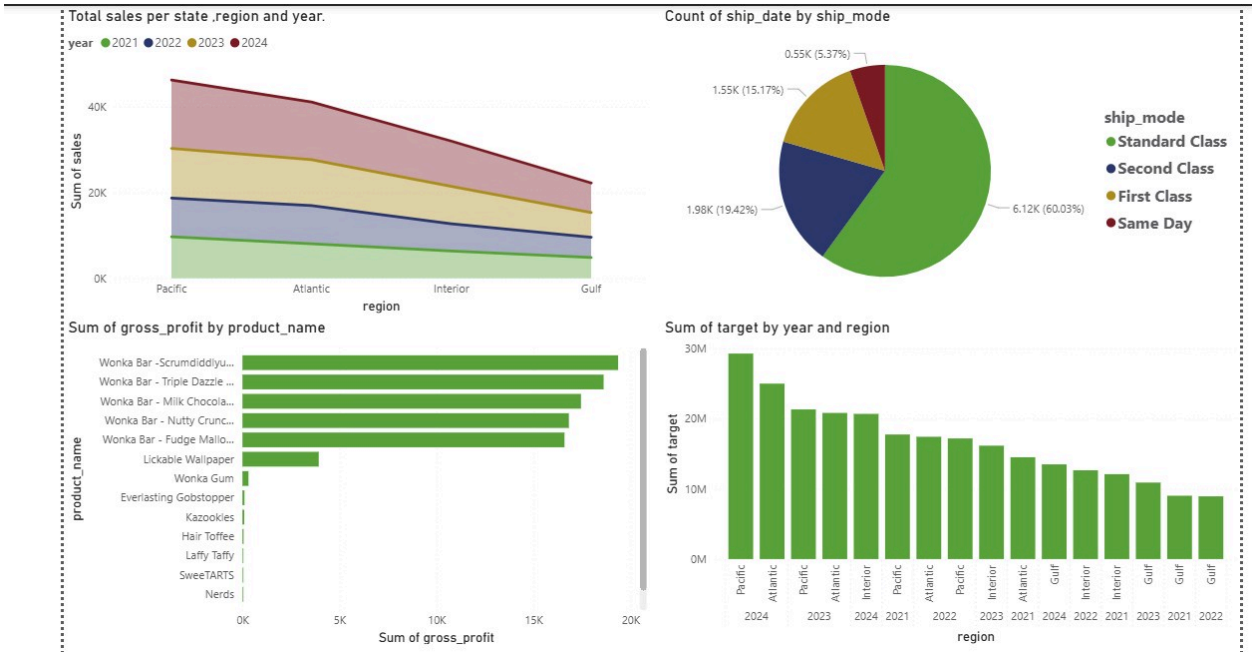
7. PowerBI Dashboard:

In this Dashboard 3 Pages were created .

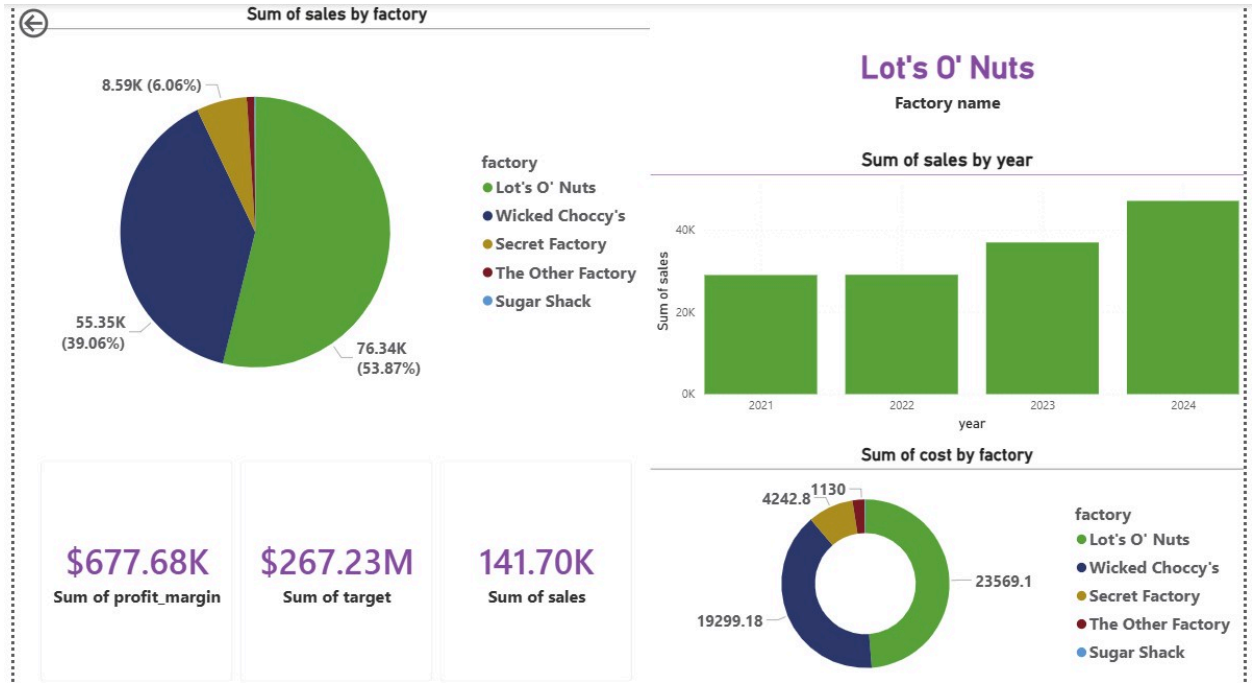
1. **Overall Information.**
2. **Region Wise.**
3. **Factory.**



<u>Overall Information of Candy Sales.</u>



Region Wise and Shipment Details.



Factory Wise Production.

8. Conclusions and Recommendations

- **Strengths:** Chocolate accounts for 92% of sales with 67% margins. Overall growth: +62% since 2021.
- **Weaknesses:** Sugar <1% of target; low volume. Other has lower margins.
- **Opportunities:** Pacific/Atlantic regions strong; expand in Interior/Gulf. Top factories (Lot's O' Nuts) scalable.
- **Recommendations:**

1. Boost Sugar via targeted marketing in low-sales regions.
2. Optimize Chocolate supply chain.
3. Investigate ship date anomalies (e.g., delays).
4. Enhance geo-analysis with US Zips for demographics.
5. Forecast 2025 using numpy linear regression on trends.

Thank You !